

Experiment # 12

Implementation of Round Robin (RR) Scheduling Algorithm

Objective

In this lab we will learn the implementation of Round Robin Scheduling Algorithm.

Software Tools

- ◆ Ubuntu 16.04
- ◆ GCC Compiler

Theory

Round Robin

- The round-robin (RR) scheduling algorithm is designed especially for time-sharing systems.
- It is similar to FCFS scheduling, but pre-emption is added to switch between processes.
- A small unit of time called time slice or quantum is defined.
- All run able processes are kept in a circular queue. The CPU scheduler goes around this queue, allocating the CPU to each process for a time interval of one quantum. New processes are added to the tail of the queue.
- Every time a process is granted the CPU, a context switch occurs, which adds overhead to the process execution time.

Performance

- q large $\Rightarrow$ FCFS
- q small $\Rightarrow$ q must be large with respect to context switch, otherwise overhead is too high.

Round Robin Scheduling Algorithm

- Get total no. of processes from user.
- Get Burst Time for each process.
- Get the Quantum/Time Slice.
- Maintain a circular ready queue.
- Make the CPU scheduler goes around the ready queue allocating CPU to each process for the time slice specified.
- Make the CPU scheduler picks the first process and after quantum expired, dispatches the process.
 - If the process have burst less than the time quantum then:
 - The process releases the CPU.

- The scheduler will then proceed to the next process in the ready queue.
- If the CPU burst of the currently running process is longer than 1 time quantum then:
 - The process will be executed for quantum time and then the process will be put at the tail of the ready queue.
 - The CPU scheduler will then select the next process in the ready queue.
- Compute Waiting Time for each process.
- Compute Turnaround Time for each process.
- Compute Average Waiting Time of processes.
- Compute Average Turnaround Time of processes.
- Display the Gantt chart.

Output:

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Enter number of processes:3
Enter burst time and priorities of processes
Burst Time for Process P1:5
Enter Priority of process P1:3
Burst Time for Process P2:7
Enter Priority of process P2:1
Burst Time for Process P3:9
Enter Priority of process P3:2
p2: burst time:7      priority:1
p3: burst time:9      priority:2
p1: burst time:5      priority:3

Process Details
  Process No.   burst time   priority   waiting time   turnaround time
  P2            7            1           0              7
  P3            9            2           7             16
  P1            5            3          16             21

Gantt Chart is as follows

      |      P2      |      P3      |      P1      |
      0              7             16             21

Average waiting time is:7.66667
Average turnaround time is:7
Finish Time is:21

```

LAB TASK

- Consider the following set of processes that arrive at time 0, with the length of the CPU burst given in the milliseconds and time quantum taken as 4 milliseconds for RR scheduling:

Process ID	Burst Time
P1	24
P2	3
P3	3

- Consider the set of 5 processes whose arrival time and burst time are given below:

Process ID	Arrival Time	Burst Time
P1	0	5
P2	1	3
P3	2	1
P4	3	2
P5	4	3

If the scheduling policy is RR with the time quantum w units, calculate average waiting time and average turnaround time.

Lab Report Evaluation:

Submit the Lab Report by writing both tasks which must include the following:

1. Procedure (Gantt Chart, average waiting time and average turnaround time calculation and algorithm/flow chart).
2. The code
3. The results (Screenshot)